

Submission Summary

Conference Name	International Conference on Data Science and AI for Social Good and Responsible Innovation	
Paper ID	129	
Paper Title	Agentic AI for Early Detection of Lung Cancer in Non-Smokers	
Abstract	Early lung cancer detection in never-smokers is an urgent unmet clinical need since its presentation is usually non-specific and there is no guideline for screening. We propose Agentic AI Framework for Explainable Early Lung Cancer Detection in Never-Smokers (ENDN) EFP that combines clinical notes and genomic data for explainable risk pro- filing. This system is based on: a Symptom Extractor Agent which retrieves pertinent symptoms and temporal information from clinical notes, a Symptom-Gene Mapper Agent which relates extracted symptoms to gene expression markers, and a Patient-Gene Scoring Agent which integrates multi-modal data to compute personalized risk scores. The system generates actionable reports for clinicians to facilitate early diagnosis. By leveraging knowledge graph reasoning, the system enables explainability and trust for AI-driven medical decisions. The system was tested on a synthetic data set containing 4,040 never-smoker patient records and the best learned model achieved 85.7% accuracy, 88.3% sensitivity, and 0.867 ROC-AUC, which surpassed baselines by 10-20%. This approach generates interpretable, evidence-based risk profiles to facilitate better outcomes for never-smoker lung cancer patients.	
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Submission Files

Agentic AI for Early Detection of Lung Cancer .pdf (315.4 Kb, 11/16/2025, 11:08:25 AM)
